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Department of Artificial Intelligence and Data Science
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AD23532 - Principles of Data Science

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Ex 4: Information Retrieval using NLTK

Name: Nischal **Roll Number:** 241801188 **Date:** 06-10-2026

Dataset: Amazon Fine Food Reviews (Kaggle) **Tools:** Python, Pandas, NumPy, NLTK, scikit-learn

Learning Objective

To understand the text preprocessing pipeline and implement a basic information retrieval system using TF-IDF vectorization, Cosine Similarity and NLTK.

Problem Statement

An e-commerce company receives thousands of customer reviews every day. Customer support executives need to retrieve reviews similar to a search query (for example "great product with fast shipping" or "poor quality") to quickly understand customer feedback. As an NLP Engineer, the task is to preprocess the review text, convert the reviews into TF-IDF feature vectors, and build a simple information retrieval system that returns the most relevant reviews for a query using Cosine Similarity.

Task 1: Load the Amazon Fine Food Reviews dataset

In [1]:

```
import pandas as pd

df = pd.read_csv('Reviews.csv')
print("Dataset loaded successfully")
print("Shape  :", df.shape)
print("Columns:")
print("   ", ", ".join(df.columns[:5]))
print("   ", ", ".join(df.columns[5:]))
```

Out [1]:

```
Dataset loaded successfully
Shape  : (568454, 10)
Columns:
    Id, ProductId, UserId, ProfileName, HelpfulnessNumerator
    HelpfulnessDenominator, Score, Time, Summary, Text
```

> **Observation:** The Amazon Fine Food Reviews file has 568,454 reviews and 10 columns. The review text is in the Text column, which is the only column used in this experiment.

Task 2: Install and import the required Python libraries

In [2]:

```
# Install (run once in a terminal / notebook):  pip install pandas numpy scikit-learn nltk
import re
import textwrap
import numpy as np
import pandas as pd
import nltk
from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics.pairwise import cosine_similarity

print("Libraries imported successfully")
```

Out [2]:

```
Libraries imported successfully
```

Task 3: Import the required NLTK resources

In [3]:

```
nltk.download('punkt', quiet=True)
nltk.download('punkt_tab', quiet=True)
nltk.download('stopwords', quiet=True)
stop_words = set(stopwords.words('english'))
print("NLTK resources ready (punkt tokenizer and English stopwords)")
```

Out [3]:

```
NLTK resources ready (punkt tokenizer and English stopwords)
```

Task 4: Clean the dataset

In [4]:

```
# Select the Text column
data = df[['Text']].copy()
print("Reviews before cleaning     :", len(data))

# Remove missing / null values
print("Missing (null) reviews      :", data['Text'].isnull().sum())
data = data.dropna(subset=['Text'])

# Remove duplicate reviews
print("Duplicate reviews           :", data.duplicated().sum())
data = data.drop_duplicates(subset='Text')

# Retain only the first 10,000 reviews
data = data.head(10000).reset_index(drop=True)
print("Reviews retained            :", len(data))
```

Out [4]:

```
Reviews before cleaning     : 568454
Missing (null) reviews      : 0
Duplicate reviews           : 174875
Reviews retained            : 10000
```

> **Observation:** No review has a missing Text value, but 174,875 reviews (about 30.8%) are exact duplicates and were removed, leaving 393,579 unique reviews. Duplicates are removed before taking the first 10,000, so the sample contains no repeated review.

Task 5: Create the preprocessing function

In [5]:

```
def preprocess(text):
    text = text.lower()                              # 1. lowercase
    text = re.sub(r'[^a-z\s]', ' ', text)            # 2. remove punctuation, numbers, special characters
    tokens = word_tokenize(text)                     # 3. tokenize
    tokens = [w for w in tokens if w not in stop_words]   # 4. remove English stopwords
    return ' '.join(tokens)                          # 5. join tokens into one string

print(preprocess("The 2 cookies were GREAT!!! Don't miss them :)"))
```

Out [5]:

```
cookies great miss
```

> **Observation:** The test sentence shows every step: "The 2 cookies were GREAT!!! Don't miss them :)" becomes "cookies great miss". The number, punctuation and emoticon are gone, and the stopwords "the", "were", "them" and the pieces "don" and "t" of the contraction are removed.

Task 6: Apply the function and create Cleaned_Text

In [6]:

```
data['Cleaned_Text'] = data['Text'].apply(preprocess)
print("Cleaned_Text column created for", len(data), "reviews")
print("Empty cleaned reviews :", (data['Cleaned_Text'] == '').sum())
print("Cleaned reviews containing the token 'br' (leftover <br /> HTML tags):",
      data['Cleaned_Text'].str.split().apply(lambda t: 'br' in t).sum())
```

Out [6]:

```
Cleaned_Text column created for 10000 reviews
Empty cleaned reviews : 0
Cleaned reviews containing the token 'br' (leftover <br /> HTML tags): 2464
```

> **Observation:** All 10,000 reviews were cleaned and none became empty. Note that 2,464 cleaned reviews (24.6%) contain the token "br": it is left over from the HTML line breaks (
) in the raw text, because removing punctuation turns
 into the word "br". It is noise, and an extra step to strip HTML tags before cleaning would remove it.

Task 7: Display original and cleaned versions of five reviews

In [7]:

```
for i in [0, 1, 2, 3, 4]:
    print(f"Review {i}")
    print(textwrap.fill("Original : " + data['Text'][i][:250] + " ...", width=105, subsequent_indent='           '))
    print(textwrap.fill("Cleaned  : " + data['Cleaned_Text'][i][:250] + " ...", width=105, subsequent_indent='           '))
    print()
```

Out [7]:

```
Review 0
Original : I have bought several of the Vitality canned dog food products and have found them all to be
           of good quality. The product looks more like a stew than a processed meat and it smells
           better. My Labrador is finicky and she appreciates this product bette ...
Cleaned  : bought several vitality canned dog food products found good quality product looks like stew
           processed meat smells better labrador finicky appreciates product better ...

Review 1
Original : Product arrived labeled as Jumbo Salted Peanuts...the peanuts were actually small sized
           unsalted. Not sure if this was an error or if the vendor intended to represent the product as
           "Jumbo". ...
Cleaned  : product arrived labeled jumbo salted peanuts peanuts actually small sized unsalted sure error
           vendor intended represent product jumbo ...

Review 2
Original : This is a confection that has been around a few centuries.  It is a light, pillowy citrus
           gelatin with nuts - in this case Filberts. And it is cut into tiny squares and then liberally
           coated with powdered sugar.  And it is a tiny mouthful of heaven.  ...
Cleaned  : confection around centuries light pillowy citrus gelatin nuts case filberts cut tiny squares
           liberally coated powdered sugar tiny mouthful heaven chewy flavorful highly recommend yummy
           treat familiar story c lewis lion witch wardrobe treat seduces ed ...

Review 3
Original : If you are looking for the secret ingredient in Robitussin I believe I have found it.  I got
           this in addition to the Root Beer Extract I ordered (which was good) and made some cherry
           soda.  The flavor is very medicinal. ...
Cleaned  : looking secret ingredient robitussin believe found got addition root beer extract ordered good
           made cherry soda flavor medicinal ...

Review 4
Original : Great taffy at a great price.  There was a wide assortment of yummy taffy.  Delivery was very
           quick.  If your a taffy lover, this is a deal. ...
Cleaned  : great taffy great price wide assortment yummy taffy delivery quick taffy lover deal ...
```

> **Observation:** Lowercasing, punctuation removal and stopword removal shorten each review while keeping the content words (for example "bought several vitality canned dog food products ..."). Stopword removal also drops "not", so "Not sure if this was an error" in Review 1 becomes "sure error". Negation is lost, which matters later for retrieval.

Task 8: Convert the cleaned reviews into TF-IDF feature vectors

In [8]:

```
tfidf = TfidfVectorizer()
tfidf_matrix = tfidf.fit_transform(data['Cleaned_Text'])

print("Shape of the TF-IDF matrix       :", tfidf_matrix.shape)
print("Total unique terms (vocabulary)  :", len(tfidf.vocabulary_))
print("Matrix density                   : %.4f%%" % (tfidf_matrix.nnz / (tfidf_matrix.shape[0]*tfidf_matrix.shape[1]) * 100))
```

Out [8]:

```
Shape of the TF-IDF matrix       : (10000, 18523)
Total unique terms (vocabulary)  : 18523
Matrix density                   : 0.1742%
```

> **Observation:** The TF-IDF matrix has 10,000 rows (reviews) and 18,523 columns (unique terms). It is very sparse (only 0.17% of cells are non-zero), as each review uses a small part of the vocabulary. TF-IDF gives a high weight to terms that are frequent in a review but rare across reviews.

Task 10: Document retrieval function (Top 5 by Cosine Similarity)

In [9]:

```
def retrieve(query, top_n=5):
    cleaned_query = preprocess(query)                           # same preprocessing as the reviews
    query_vec = tfidf.transform([cleaned_query])                # query -> TF-IDF vector
    scores = cosine_similarity(query_vec, tfidf_matrix).ravel()  # similarity with every review
    top_idx = scores.argsort()[::-1][:top_n]                    # indices of the Top 5

    print(f'Query         : "{query}"')
    print(f'Cleaned query : "{cleaned_query}"')
    print("-" * 105)
    for rank, i in enumerate(top_idx, start=1):
        print(f"Rank {rank} | Similarity score: {scores[i]:.4f} | Review index: {i}")
        print(textwrap.fill("Original : " + data['Text'][i], width=105, subsequent_indent='           '))
        print(textwrap.fill("Cleaned  : " + data['Cleaned_Text'][i], width=105, subsequent_indent='           '))
        print()
    return top_idx, scores[top_idx]

print("Retrieval function defined")
```

Out [9]:

```
Retrieval function defined
```

> **Observation:** The same preprocess() function is applied to the query so that it uses the same vocabulary as the reviews. The query is converted with the already fitted vectorizer (transform, not fit_transform), and cosine similarity measures the angle between the query vector and each review vector, so longer reviews are not favoured by length.

Task 11.1: Test query 1

In [10]:

```
_ = retrieve("great product with fast shipping")
```

Out [10]:

```
Query         : "great product with fast shipping"
Cleaned query : "great product fast shipping"
-----------------------------------------------------------------------------------------------------
Rank 1 | Similarity score: 0.5103 | Review index: 5210
Original : Enjoyed the product and they also provided very fast shipping.  I'm about out and need to
           order more.
Cleaned  : enjoyed product also provided fast shipping need order

Rank 2 | Similarity score: 0.4925 | Review index: 7992
Original : The tea is good and fresh. We enjoy it. The shipping is fast and cost is reasonable.<br />I
           recommend this product.
Cleaned  : tea good fresh enjoy shipping fast cost reasonable br recommend product

Rank 3 | Similarity score: 0.4001 | Review index: 5041
Original : These are very good, they were a great price and fast-free shipping when you purchase enough
           food.  The only thing is that the box was smaller than I guessed it would be.
Cleaned  : good great price fast free shipping purchase enough food thing box smaller guessed would

Rank 4 | Similarity score: 0.3996 | Review index: 6011
Original : The energy drink is a great product. The shipping price is crazy tho. I paid more for the
           shipping then both products together, so not a great deal.
Cleaned  : energy drink great product shipping price crazy tho paid shipping products together great deal

Rank 5 | Similarity score: 0.3936 | Review index: 3902
Original : Was exactly what I was looking for. Super fast shipping! A bit on the spicy side,but great
           taste!!!!<br />Would definately purchase them again!
Cleaned  : exactly looking super fast shipping bit spicy side great taste br would definately purchase
```

> **Observation:** The query is cleaned to "great product fast shipping". Ranks 1, 2, 3 and 5 are relevant: they praise the product and the fast shipping, and the top two scores (0.51 and 0.49) are clearly the highest. Rank 4 matches on the words "great", "product" and "shipping", but the reviewer complains about the shipping price ("not a great deal"), so it is a lexical match with the wrong sentiment.

Task 11.2: Test query 2

In [11]:

```
_ = retrieve("disappointed")
```

Out [11]:

```
Query         : "disappointed"
Cleaned query : "disappointed"
-----------------------------------------------------------------------------------------------------
Rank 1 | Similarity score: 0.4846 | Review index: 3140
Original : I am a bit disappointed.  The flavor was not what I wanted or expected.
Cleaned  : bit disappointed flavor wanted expected
Rank 2 | Similarity score: 0.3582 | Review index: 4365
```

```
Original : The product is very good. Way too expensive and at almost 5.00 a box to get pancakes or
           biscuits I was disappointed
Cleaned  : product good way expensive almost box get pancakes biscuits disappointed

Rank 3 | Similarity score: 0.3216 | Review index: 785
Original : I was disappointed in this product because I thought it would be bigger.<br />Also, it did not
           come with enough icing. I had to use my own.
Cleaned  : disappointed product thought would bigger br also come enough icing use

Rank 4 | Similarity score: 0.3195 | Review index: 6523
Original : Disappointed.  The big boxes had a very different flavor than the smaller boxes.  I am not
           sure why but not good.
Cleaned  : disappointed big boxes different flavor smaller boxes sure good

Rank 5 | Similarity score: 0.3117 | Review index: 3322
Original : Just plain nasty!!! This item tasted like cardboard, very watered down and no flavor to it at
           all.  Very disappointed
Cleaned  : plain nasty item tasted like cardboard watered flavor disappointed
```

Observation: All five results express disappointment. Rank 1 (0.48) is a short review dominated by the word, and the scores fall quickly to about 0.31 by rank 5. Rank 2 praises the product ("very good") and is disappointed only about the price, so the retrieved reviews differ in what they are disappointed about.

Task 11.3: Test query 3

In [12]:

```
_ = retrieve("excellent taste")
```

Out [12]:

```
Query         : "excellent taste"
Cleaned query : "excellent taste"
--------------------------------------------------------------------------------------------------------------
Rank 1 | Similarity score: 0.5020 | Review index: 2326
Original : It is very very good sweets<br />Amazon always is excellent<br />Amazon always is excellent<br
           />Amazon always is excellent<br />Amazon always is excellent
Cleaned  : good sweets br amazon always excellent br amazon always excellent br amazon always excellent
           br amazon always excellent

Rank 2 | Similarity score: 0.4605 | Review index: 1740
Original : The Popchips were excellent - had excellent flavor and low in calories.  When you eat these -
           don't need another potato chip!
Cleaned  : popchips excellent excellent flavor low calories eat need another potato chip

Rank 3 | Similarity score: 0.4054 | Review index: 7298
Original : Excellent product. Exactly what they said it would be. Superior chamomile from what I have
           ordered before from other companies. Excellent value for the price.
Cleaned  : excellent product exactly said would superior chamomile ordered companies excellent value
           price

Rank 4 | Similarity score: 0.3856 | Review index: 3396
Original : We recieved the product on time and in excellent shape. It tastes good and meets our
           expectations. Thanks for the excellent service.
Cleaned  : recieved product time excellent shape tastes good meets expectations thanks excellent service

Rank 5 | Similarity score: 0.3840 | Review index: 9471
Original : Title says it all, a Excellent quality product and Excellent service at a fair price. Product
           came ahead of time and as expected.
Cleaned  : title says excellent quality product excellent service fair price product came ahead time
           expected
```

Observation: Results are only partly relevant. Rank 1 (0.50) ranks first because the word "excellent" is repeated four times, yet it is about Amazon's service and not taste. Only rank 2 ("excellent flavor") and rank 4 ("tastes good") speak about taste. The query word "taste" differs from "tastes" and "flavor", and TF-IDF has no stemming or synonyms, so these are not matched with it.

Task 11.4: Test query 4

In [13]:

```
_ = retrieve("poor packaging")
```

Out [13]:

```
Query         : "poor packaging"
Cleaned query : "poor packaging"
--------------------------------------------------------------------------------------------------------------
Rank 1 | Similarity score: 0.3177 | Review index: 6162
Original : I was delighted to find this product online, because I cannot always find it in the stores.
           However, each package I have opened thus far is at least half reduced to crumbs because of
           poor packaging for shipping.  I doubt that I will order more using this method.
Cleaned  : delighted find product online cannot always find stores however package opened thus far least
           half reduced crumbs poor packaging shipping doubt order using method
Rank 2 | Similarity score: 0.3085 | Review index: 4595
```

```
Original : This review is for the packaging and not the actual mix, since the mix is a very subjective
           thing and my taste my not match others and, some wrote about how poor the packaging was. All I
           can say is with mine there is more than enough labeling, on the bags themselves and on the box
           there are instructions on how to. Granted the box is nothing heavy duty or special and the mix
           is in plain white bags, there are more than enough instructions to go by.
Cleaned  : review packaging actual mix since mix subjective thing taste match others wrote poor packaging
           say mine enough labeling bags box instructions granted box nothing heavy duty special mix
           plain white bags enough instructions go

Rank 3 | Similarity score: 0.3083 | Review index: 7553
Original : The soup is rich and tasty, but DON'T buy it from Amazon.  The packaging is so poor that the
           cans arive dented as did the replacement shipment. I got my money back but now have to pay
           more to buy this wonderful soup from my local grocer.
Cleaned  : soup rich tasty buy amazon packaging poor cans arive dented replacement shipment got money
           back pay buy wonderful soup local grocer

Rank 4 | Similarity score: 0.2920 | Review index: 2002
Original : Didnt really enjoy these after wanting to try something other than aunt jemima's from
           Dominicks. The pancakes tasted strange and sometimes forgettable, but it might have been due
           to the poor packaging they came in along with the syrup.
Cleaned  : didnt really enjoy wanting try something aunt jemima dominicks pancakes tasted strange
           sometimes forgettable might due poor packaging came along syrup

Rank 5 | Similarity score: 0.2811 | Review index: 9663
Original : was really dissappointed in product... too salty and poor flavor....Convenience is what makes
           product worth purchasing if you like pork rinds...
Cleaned  : really dissappointed product salty poor flavor convenience makes product worth purchasing like
           pork rinds
```

> **Observation:** This query has the lowest scores (0.28 to 0.32). Ranks 1 to 4 are relevant complaints about packaging (broken, dented or poorly packed products). Rank 5 matches only the word "poor" in "poor flavor", so it is not about packaging. The scores are lower because the reviews that mention packaging are longer, and cosine similarity divides by review length.

Task 12: Compare the retrieved reviews for different queries

Summary of the four queries

Query	Score range (Top 5)	Relevant results	Main issue
great product with fast shipping	0.39 - 0.51	4 of 5	Rank 4 matches the words but is negative about shipping cost
disappointed	0.31 - 0.48	5 of 5	Single-word query; reasons for disappointment vary
excellent taste	0.39 - 0.50	2 of 5 about taste	Repeated "excellent" outranks taste; no stemming or synonyms
poor packaging	0.28 - 0.32	4 of 5	Rank 5 matches only "poor"; scores lower for longer reviews

> **Observation:** The system works best for distinctive, specific words ("disappointed", "packaging", "shipping"), where top results are clearly on topic. It is weaker where the meaning needs more than word matching: it does not understand synonyms ("taste" vs "flavor"), word forms ("taste" vs "tastes"), or sentiment and negation (the stopword "not" is removed, and a negative review about shipping cost matched a positive shipping query). Scores are higher for short reviews that repeat the query words and lower for longer reviews, because cosine similarity normalises by length. Possible improvements are stemming or lemmatization, removing HTML tags such as "br", keeping negation words, and using bigrams in TF-IDF.

Result

Thus a basic information retrieval system was built on the Amazon Fine Food Reviews dataset. After removing 174,875 duplicate reviews, the first 10,000 unique reviews were cleaned (lowercasing, removal of punctuation, numbers and special characters, tokenization and stopword removal using NLTK) and converted into a TF-IDF matrix of shape (10000, 18523). A retrieval function that preprocesses the query, vectorizes it and ranks reviews by cosine similarity returned the Top 5 reviews for the four test queries. The results were mostly relevant for specific words but limited by exact word matching, as the comparison of the queries showed.